

Rohan Jain

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SUMMARY

MS Data Science graduate from CU Boulder (GPA 3.9/4.0, May 2026) with production experience building and hardening LLM agent systems as an AI Engineering Intern at Bright Machines - LangGraph agents, MCP tool servers, Kubernetes-deployed AI infrastructure. Enforces schema-level reliability on LLM outputs and ships end-to-end AI systems under real production constraints. Seeking AI Engineer or applied LLM roles focused on reliability and generative AI.

SKILLS

LLM & Agents: Python, LangGraph (ReAct, multi-agent), LangChain, Model Context Protocol (MCP; built and hardened MCP servers), LLM tool-calling agents, RAG (BM25, embeddings, hybrid), Pydantic v2 structured outputs, LiteLLM multi-provider gateway, OpenAI-compatible APIs, OpenAI / Anthropic / Gemini APIs, Ollama, LLM evaluation harnesses, prompt-injection testing, guardrails, Langfuse tracing; agentic coding tools (Claude Code, Cursor)

Backend & Data: FastAPI, REST APIs, PostgreSQL + Alembic, MySQL, SQL, SQLite / sqlite-vec, FAISS, PyTorch, TensorFlow, scikit-learn, Databricks (Unity Catalog, Unity AI Gateway, Model Serving, Genie), Temporal, Docker, Kubernetes / Kustomize, ArgoCD, GitLab CI/CD, GitHub Actions, pytest

Frontend, Cloud & Observability: TypeScript, React, Next.js; Azure (AKS, ACR, Entra ID), AWS (Lambda, S3, EC2, API Gateway, RDS); OIDC auth (oauth2-proxy, Keycloak); Grafana, Loki, Prometheus, MLflow

WORK EXPERIENCE

AI Engineering Intern - Bright Machines

Jun 2026 – Present | San Francisco, CA

- Owned tool selection for an on-prem LLM agent (**LangGraph** ReAct over **LiteLLM**): 132 **MCP** tool schemas exceeded the 32K-token window; built a 115-question harness that raised recall 0.47 to 0.97.
- Designed and shipped a read-only-by-construction **MCP** server from scratch (37 files, +3,163 lines, 12 tools) serving a 32,000-workflow **Temporal** namespace in production; Docker-packaged, deployed via **Kubernetes/Kustomize**.
- Hardened four **MCP** servers behind a production LLM agent: closed a Critical path-traversal vulnerability (**CWE-22**) with 9 regression tests, removed cluster-secret access from a **Kubernetes** service account, enforced least-privilege RBAC.
- Root-caused a **74-minute** silent chat outage (agent registry not rehydrating after a gateway restart while health probes stayed green), disproved the planned upgrade fix by inspecting the running container, and shipped **Postgres** persistence plus a self-healing watchdog (**135 tests**), verified live in production.
- Built an on-prem grounded document Q&A platform end to end: **FastAPI** backend, **React/TypeScript** UI, sqlite-vec retrieval, cited answers, and an admin sync worker; **352 automated tests** across 47 commits.
- Built Confluence knowledge pipeline to run fully on-prem (**CQL** fetch in code, **Qwen** on an H100, no data egress) instead of per-page cloud LLM calls estimated at \$6/page, avoiding over **\$12K** of API spend across a 2,000+ page estate; recurring cost tracks ~12 changed pages/month.
- Led the agent-orchestration architecture spike for a production **LangGraph** agent: evaluated the **A2A** (agent-to-agent) registry protocol against Temporal and **DBOS** as a durable backbone, showed transport is not durability, and published a stick/adopt/hybrid roadmap to Confluence for team review.

Data Science & Engineering Intern - Nexus Weather & Climate

May 2025 – May 2026 | Boulder, CO

- Replaced legacy SVM bias-correction with a tuned Random Forest for NWP post-processing; applied feature engineering and cross-validation, reducing temperature MAE 47% and humidity MAE 54%.
- Implemented an automated inference quality gate rejecting models whose post-ML MAE fails to beat the pre-ML baseline; reduced failed production runs 90% and saved 20 engineer-hours monthly.

Software Developer - Kopf Lab, University of Colorado Boulder

Dec 2025 – May 2026 | Boulder, CO

- Reverse-engineered proprietary Thermo Isodat binary formats via DLL decompilation, building deterministic R readers; migrated lab infrastructure to AWS EC2 with Keycloak OIDC and Grafana/Loki/Prometheus observability.

EDUCATION

University of Colorado, Boulder

Master's in Data Science (GPA: 3.9/4.0)

Aug 2024 – May 2026 | Boulder, CO

Mumbai University, Rajiv Gandhi Institute of Technology

B.E. in Computer Engineering (GPA: 3.7/4.0)

Jul 2020 – May 2024 | Mumbai, India

PROJECTS

ClaimPilot AI - Guardrailed LLM Agent | Python, Pydantic v2, OpenAI function calling

Jan 2026

- Architected a hybrid agent for healthcare claims validation: a deterministic Pydantic layer validates codes while OpenAI function calling generates structured fixes; strict schemas with retry logic guarantee a valid report.

AgentSquared - No-Code AI Agent Builder | Python, FastAPI, SQLAlchemy, Next.js, Tailwind, Gemini API

Mar 2026

- Built a no-code AI agent platform at HackCU 12: config-driven architecture where every agent is a database row plus JSON config, Gemini-powered RAG, and real-time Bluesky brand monitoring with human-approved replies.